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# ENERGY CONVERSION-I

## EEE 221

## LECTURE-06

## TRN

Department of EEE  
Brac University

Digital Nomad Guide:

Thomas



Inspiring Excellence

5<sup>th</sup> Edition

Chapter 6

Induction Motors

# Chapter 6: Induction Motors

- Construction of Induction Motors
- Basic Induction Motor Concepts
- Equivalent circuit of an induction motor
- Power and Torque in induction motor
- Torque-speed characteristics
- Starting Induction Motors
- Speed Control of Induction Motors
- Single phase induction motors

[https://youtu.be/AQqyGNOP\\_3o](https://youtu.be/AQqyGNOP_3o)

# Induction Motor Construction

- There are two different types of induction motor rotors which can be placed inside the stator. One is caged a *cage rotor*, while the other is called a *wound rotor*.
- Two basic design types depending on the rotor design –
- **Cage-rotor**: conducting bars laid into slots and shorted at both ends by shorting rings.
- **Wound-rotor**: complete set of three-phase windings exactly as the stator. Usually Y-connected, the ends of the three rotor wires are connected to 3 slip rings on the rotor shaft. In this way, the rotor circuit is accessible.

# Induction Motor Construction

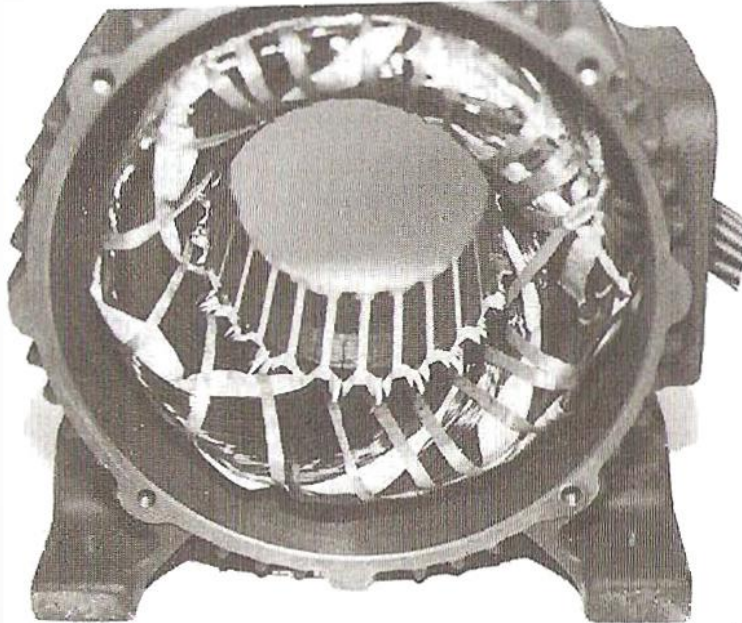
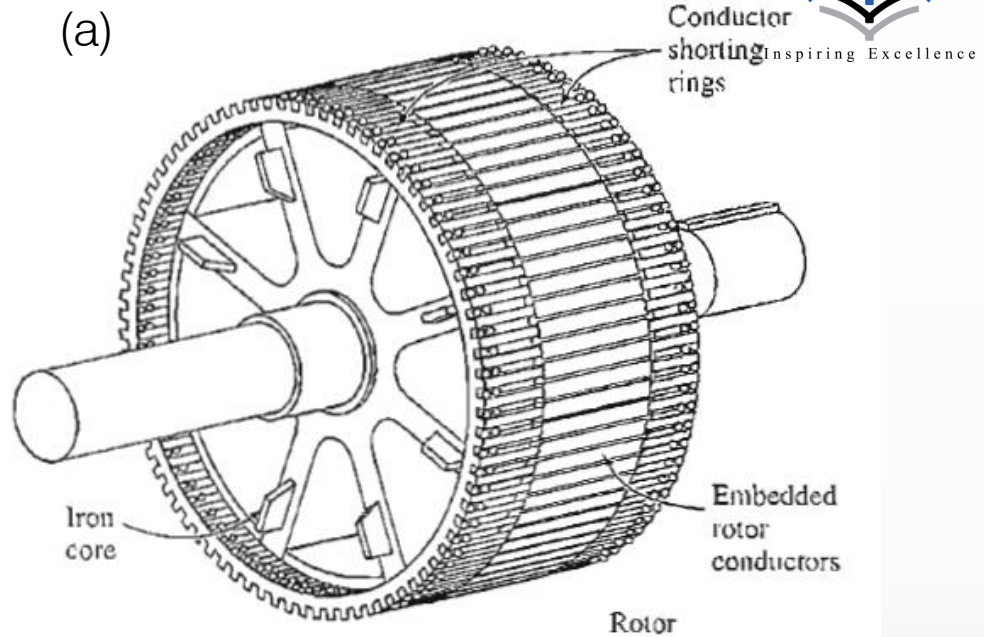


FIGURE 6-1  
The stator of a typical induction motor, showing the stator windings.



(b)

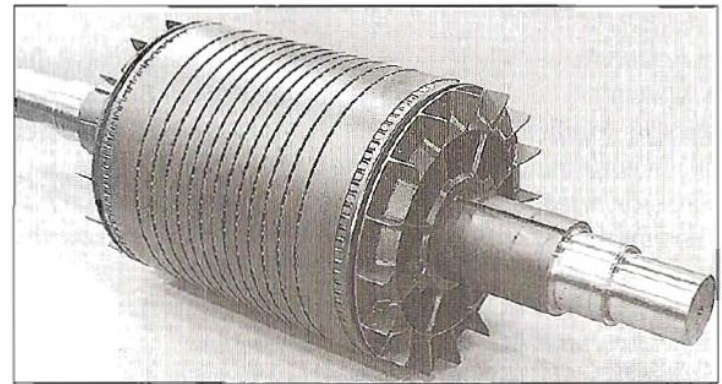


FIGURE 6-2  
(a) Sketch of cage rotor  
(b) A typical cage rotor.

# Cage rotor induction motor

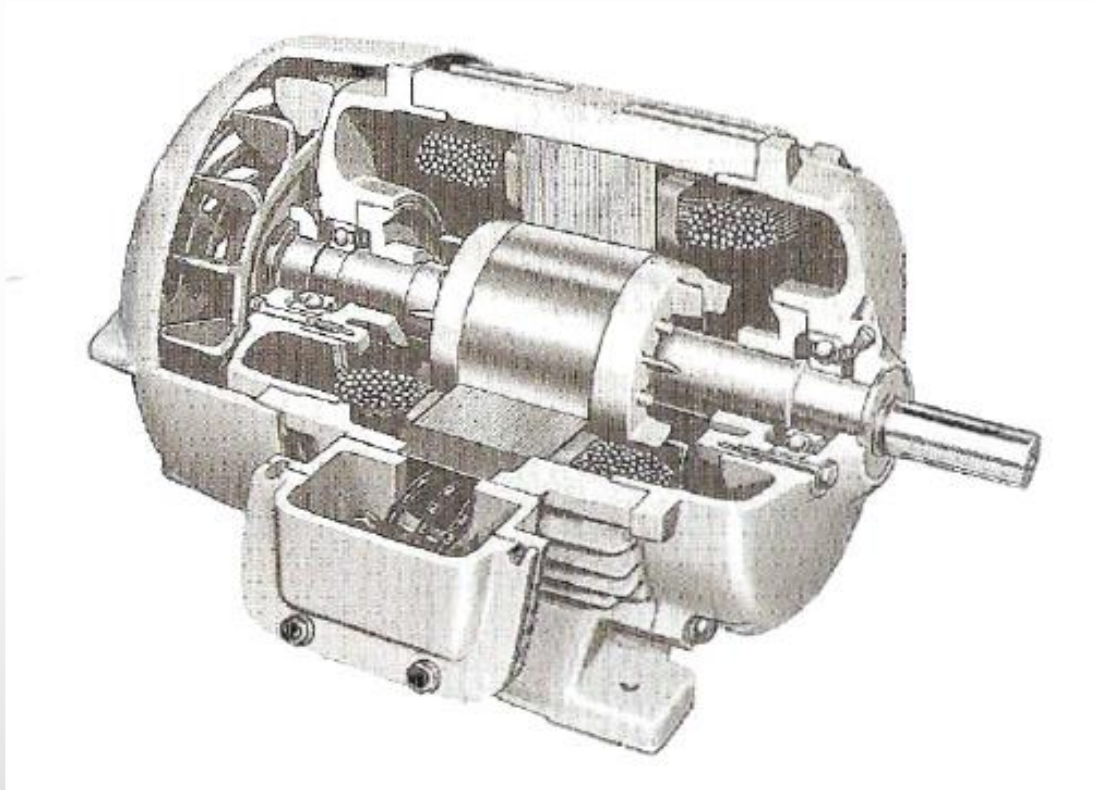


FIGURE 6-3

(a) Cutaway diagram of a typical small cage rotor induction motor

# Wound rotor of induction motors

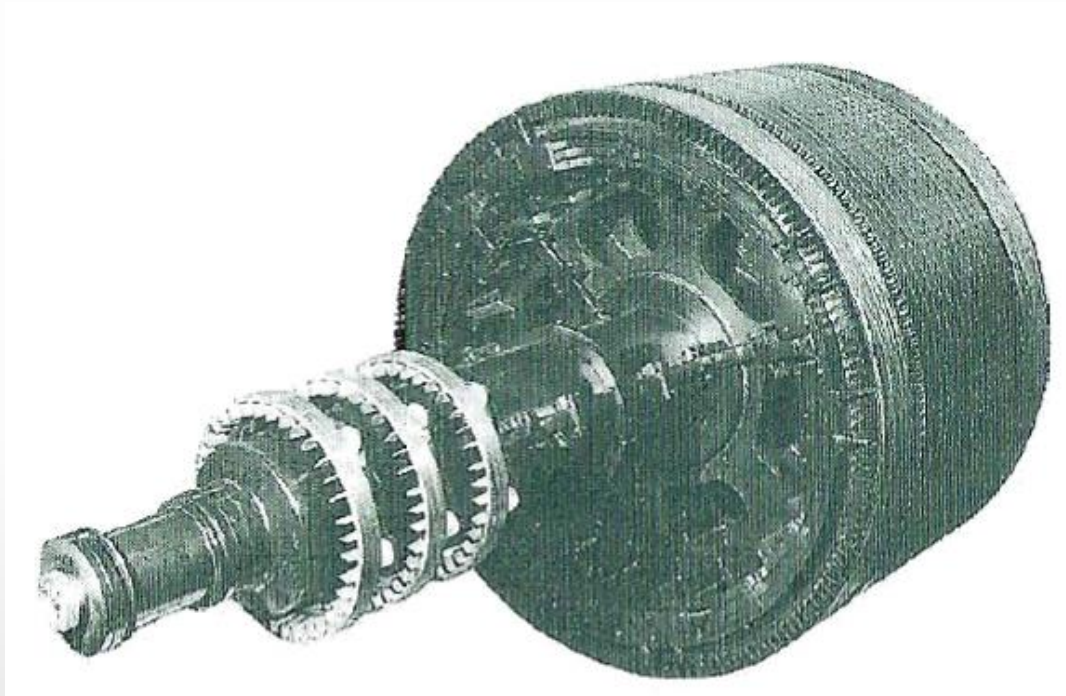


FIGURE 6-4

Typical wound rotors for induction motors. Notice the slip rings and the bars connecting the rotor windings to the slip rings



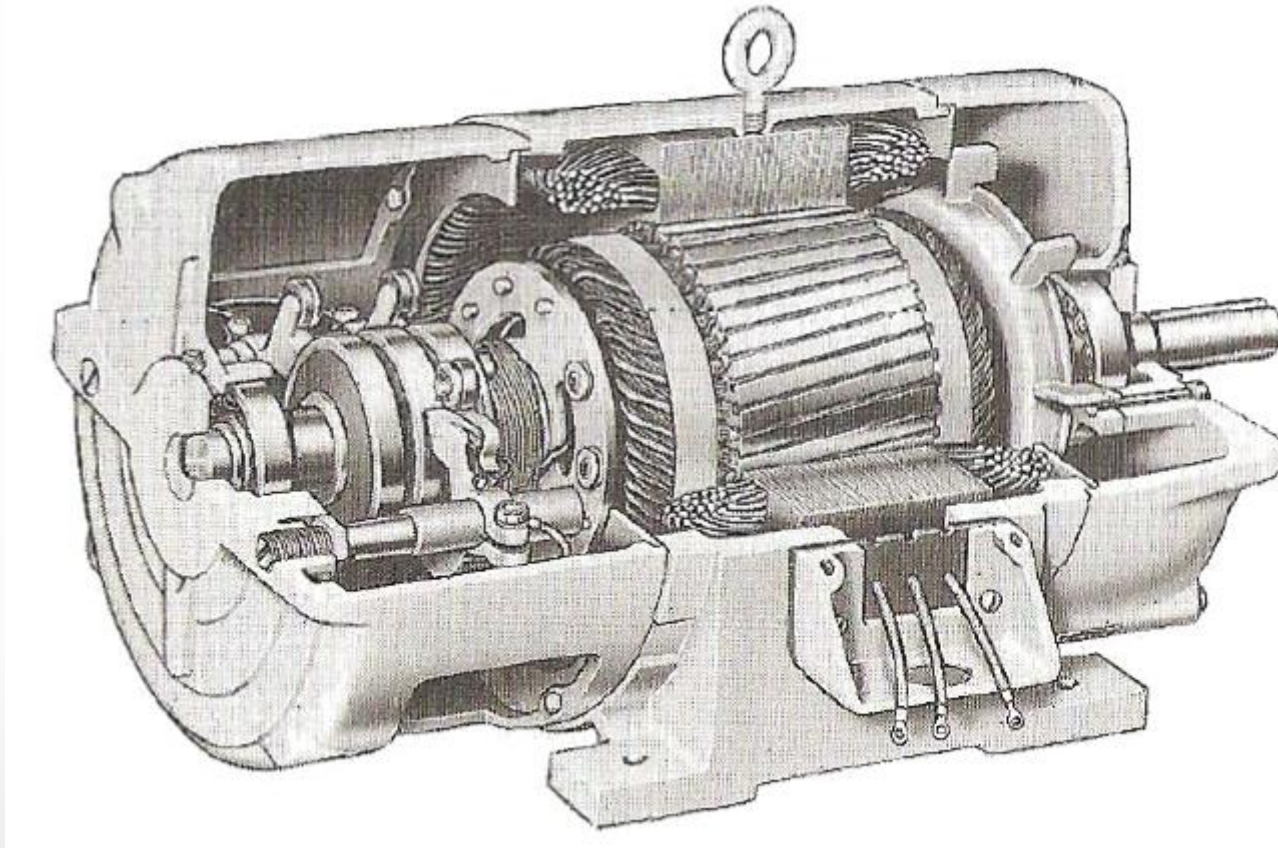


FIGURE 6-5

Cutaway diagram of a wound-rotor induction motor. Notice the brushes and slip rings. Also notice that the rotor windings are skewed to eliminate slot harmonics. (Courtesy of MagIleTek, Inc.)

# Basic induction motor concepts

Figure 6–6 shows a cage rotor induction motor. A three-phase set of voltages has been applied to the stator, and a three-phase set of stator currents is flowing. These currents produce a magnetic field  $\mathbf{B}_s$ , which is rotating in a counterclockwise direction. The speed of the magnetic field's rotation is given by

$$n_{\text{sync}} = \frac{120 f_{se}}{P} \quad (6-1)$$

where  $f_{se}$  is the system frequency applied to the stator in hertz and  $P$  is the number of poles in the machine. This rotating magnetic field  $\mathbf{B}_s$  passes over the rotor bars and induces a voltage in them.

The voltage induced in a given rotor bar is given by the equation

$$e_{\text{ind}} = (\mathbf{v} \times \mathbf{B}) \cdot \mathbf{l} \quad (1-45)$$

where  $\mathbf{v}$  = velocity of the bar *relative to the magnetic field*

$\mathbf{B}$  = magnetic flux density vector

$\mathbf{l}$  = length of conductor in the magnetic field



# The development of induced torque in an induction motor

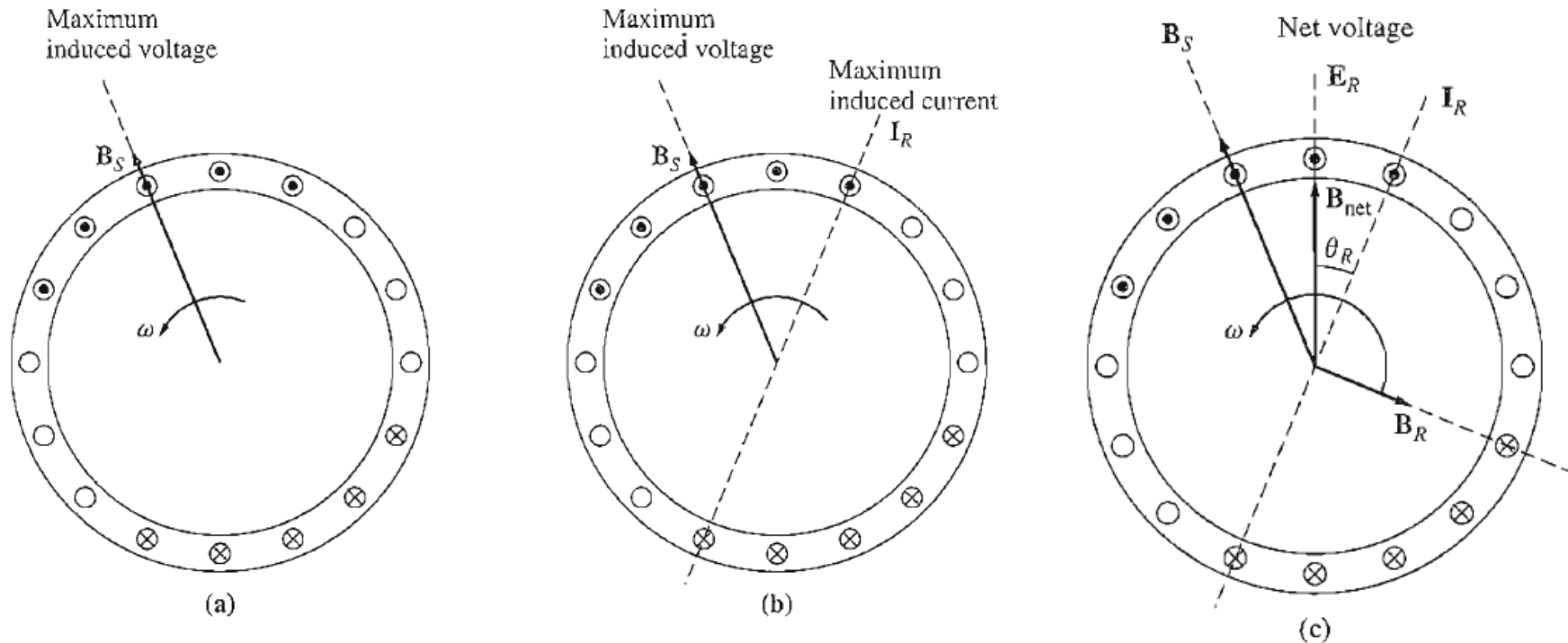


Figure 6-6

The development of induced torque in an induction motor. (a) The rotating stator field  $B_S$  induces a voltage in the rotor bars; (b) the rotor voltage produces a rotor current flow, which lags behind the voltage due to rotor inductance; (c) the rotor current produces a magnetic field  $B_R$  lagging rotor current by  $90^\circ$ . Interaction between  $B_R$  and  $B_S$  produces a torque in the machine.

It is the *relative* motion of the rotor compared to the stator magnetic field that produces induced voltage in a rotor bar. The velocity of the upper rotor bars relative to the magnetic field is to the right, so the induced voltage in the upper bars is out of the page, while the induced voltage in the lower bars is into the page. This results in a current flow out of the upper bars and into the lower bars. However, since the rotor assembly is inductive, the peak rotor current lags behind the peak rotor voltage (see Figure 6–6b). The rotor current flow produces a rotor magnetic field  $\mathbf{B}_R$ .

Finally, since the induced torque in the machine is given by

$$\tau_{\text{ind}} = k\mathbf{B}_R \times \mathbf{B}_S \quad (3-58)$$

the resulting torque is counterclockwise. Since the rotor induced torque is counterclockwise, the rotor accelerates in that direction.

There is a finite upper limit to the motor's speed, however. If the induction motor's rotor were turning at *synchronous speed*, then the rotor bars would be stationary *relative to the magnetic field* and there would be no induced voltage. If  $e_{\text{ind}}$  were equal to 0, then there would be no rotor current and no rotor magnetic field. With no rotor magnetic field, the induced torque would be zero, and the rotor would slow down as a result of friction losses. An induction motor can thus speed up to near-synchronous speed, but it can never exactly reach synchronous speed.

Note that in normal operation *both the rotor and stator magnetic fields  $\mathbf{B}_R$  and  $\mathbf{B}_S$  rotate together at synchronous speed  $n_{\text{sync}}$ , while the rotor itself turns at a slower speed.*

# The concept of Rotor Slip

The voltage induced in a rotor bar of an induction motor depends on the speed of the rotor *relative to the magnetic fields*. Since the behavior of an induction motor depends on the rotor's voltage and current, it is often more logical to talk about this relative speed. Two terms are commonly used to define the relative motion of the rotor and the magnetic fields. One is *slip speed*, defined as the difference between synchronous speed and rotor speed:

$$n_{\text{slip}} = n_{\text{sync}} - n_m \quad (6-2)$$

where  $n_{\text{slip}}$  = slip speed of the machine  
 $n_{\text{sync}}$  = speed of the magnetic fields  
 $n_m$  = mechanical shaft speed of motor

The other term used to describe the relative motion is *slip*, which is the relative speed expressed on a per-unit or a percentage basis. That is, slip is defined as

$$s = \frac{n_{\text{slip}}}{n_{\text{sync}}} (\times 100\%) \quad (6-3)$$

$$s = \frac{n_{\text{sync}} - n_m}{n_{\text{sync}}} (\times 100\%) \quad (6-4)$$

This equation can also be expressed in terms of angular velocity  $\omega$  (radians per second) as

$$s = \frac{\omega_{\text{sync}} - \omega_m}{\omega_{\text{sync}}} (\times 100\%) \quad (6-5)$$

Notice that if the rotor turns at synchronous speed,  $s = 0$ , while if the rotor is stationary,  $s = 1$ . All normal motor speeds fall somewhere between those two limits.

It is possible to express the mechanical speed of the rotor shaft in terms of synchronous speed and slip. Solving Equations (6-4) and (6-5) for mechanical speed yields

$$n_m = (1 - s)n_{\text{sync}} \quad (6-6)$$

or

$$\omega_m = (1 - s)\omega_{\text{sync}} \quad (6-7)$$

These equations are useful in the derivation of induction motor torque and power relationships. Energy Conversion 1

# The Electrical Frequency on the Rotor

An induction motor works by inducing voltages and currents in the rotor of the machine, and for that reason it has sometimes been called a *rotating transformer*. Like a transformer, the primary (stator) induces a voltage in the secondary (rotor), but *unlike* a transformer, the secondary frequency is not necessarily the same as the primary frequency.

If the rotor of a motor is locked so that it cannot move, then the rotor will have the same frequency as the stator. On the other hand, if the rotor turns at synchronous speed, the frequency on the rotor will be zero. What will the rotor frequency be for any arbitrary rate of rotor rotation?

At  $n_m = 0$  r/min, the rotor frequency  $f_{re} = f_{se}$ , and the slip  $s = 1$ . At  $n_m = n_{sync}$ , the rotor frequency  $f_{re} = 0$  Hz, and the slip  $s = 0$ . For any speed in between, the rotor frequency is directly proportional to the *difference* between the speed of the magnetic field  $n_{sync}$  and the speed of the rotor  $n_m$ . Since the slip of the rotor is defined as

$$s = \frac{n_{sync} - n_m}{n_{sync}} \quad (6-4)$$

the rotor frequency can be expressed as

$$\boxed{f_{re} = sf_{se}} \quad (6-8)$$



Several alternative forms of this expression exist that are sometimes useful. One of the more common expressions is derived by substituting Equation (6-4) for the slip into Equation (6-8) and then substituting for  $n_{\text{sync}}$  in the denominator of the expression:

$$f_{re} = \frac{n_{\text{sync}} - n_m}{n_{\text{sync}}} f_{se}$$

But  $n_{\text{sync}} = 120 f_{se} / P$  [from Equation (6-1)], so

$$f_{re} = (n_{\text{sync}} - n_m) \frac{P}{120 f_{se}} f_{se}$$

Therefore,

$$\boxed{f_{re} = \frac{P}{120} (n_{\text{sync}} - n_m)} \quad (6-9)$$

**Example 6–1.** A 208-V, 10-hp, four-pole, 60-Hz, Y-connected induction motor has a full-load slip of 5 percent.

- (a) What is the synchronous speed of this motor?
- (b) What is the rotor speed of this motor at the rated load?
- (c) What is the rotor frequency of this motor at the rated load?
- (d) What is the shaft torque of this motor at the rated load?